



Prospective assessment of aneurysmal rupture risk scores in patients with subarachnoid hemorrhage: a multicentric cohort

P. Lognon¹ · F. Gariel² · G. Marnat² · J. Darcourt³ · P. Constant dit Beaufils^{4,5} · J. Burel⁶ · E. Shotar⁷ · J. F. Hak⁸ · C. Fauché⁹ · B. Kerleroux¹⁰ · A. Guédon¹¹ · J. Ognard¹² · G. Forestier¹³ · R. Pop¹⁴ · C. Paya¹⁵ · J. B. Veyrières¹⁵ · P. Sporns^{16,17} · J. B. Girot¹⁸ · R. Zannoni¹⁹ · F. Zhu²⁰ · A. Crespy¹ · V. L'Allinec¹⁸ · D. Mihoc¹⁴ · A. Rouchaud¹³ · J. C. Gentric¹² · W. Ben Hassen¹⁰ · N. Raynaud⁹ · B. Testud⁸ · F. Clarençon⁷ · B. Kaczmarek⁶ · R. Bourcier^{4,5} · G. Bellanger³ · G. Boulouis¹ · Kevin Janot¹ · on behalf of the JENI Research Collaborative

Received: 2 March 2022 / Accepted: 20 May 2022

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2022

Abstract

Purpose The natural evolution of unruptured intracranial aneurysms (UIA) is indeed difficult to predict at the individual level.

Objective In a large prospective multicentric European cohort, we aimed to evaluate whether the PHASES, UCAS, and ELAPSS scores in patients with aneurysmal subarachnoid hemorrhage would have predicted a high risk of aneurysmal rupture or growth.

Methods Academic centers treating patients with intracranial aneurysms were invited to prospectively collect de-identified data from all patients admitted at their institution for a subarachnoid hemorrhage-related to intracranial aneurysmal rupture between January 1 and March 31, 2021 through a trainee-led research collaborative network. Each responding center was provided with an electronic case record form (CRF) which collected all the elements of the PHASES, ELAPSS, and UCAS scores.

Results A total of 319 patients with aneurysmal subarachnoid hemorrhage were included at 17 centers during a 3-month period. One hundred eighty-three aneurysms (57%) were less than 7 mm. The majority of aneurysms were located on the anterior communicating artery ($n = 131$, 41%). One hundred eighty-four patients (57%), 103 patients (32%), and 58 (18%) were classified as having a low risk of rupture or growth, according to the PHASES, UCAS, and ELAPSS scores, respectively.

Conclusion In a prospective study of European patients with aneurysmal subarachnoid hemorrhage, we showed that 3 common risk-assessment tools designed for patients with unruptured intracranial aneurysms would have not identified most patients to be at high or intermediate risk for rupture, questioning their use for decision-making in the setting of unruptured aneurysms.

Keywords Aneurysm · Subarachnoid hemorrhage

Abbreviations

ACA	Anterior cerebral artery
ACom	Anterior communicating artery
BA	Basilar artery
CAM	Comprehensive Aneurysm Management
ELAPSS	Earlier subarachnoid hemorrhage, location of the aneurysm, age > 60 years, population, size of the aneurysm, shape of the aneurysm
ICA	Internal carotid artery
JENI	Jeune en Neuroradiologie Interventionnelle

MCA	Middle cerebral artery
mRS	Modified Rankin Scale
PCA	Posterior cerebral artery
PCom	Posterior communicating artery
PHASES	Population, Hypertension, Age, Size of aneurysm, Earlier subarachnoid hemorrhage, Site of aneurysm
SAH	Subarachnoid hemorrhage
SFNR	Société Française de Neuroradiologie
UCAS	Japanese Unruptured Cerebral Aneurysm Study
UIA	Unruptured intracranial aneurysm
VA	Vertebral artery

✉ Kevin Janot
kevin.janot@univ-tours.fr

Extended author information available on the last page of the article

Introduction

The incidence of unruptured intracranial aneurysms (UIA) is estimated between 3 and 5% in the general population [1, 2], and their management remains heterogeneous [3, 4]. The natural evolution of UIA is indeed difficult to predict at the individual level, with a risk of spontaneous rupture estimated between 0.1 and 2% per year at the group level [5, 6]. As patients experiencing a rupture present very high morbidity [7], a preventive strategy for UIAs is appealing to prevent such evolution. Yet UIA treatment, whether surgical or endovascular, also exposes the patient to serious complications at rates of 3 to 10% [8, 9]. To weigh the benefits of preventive treatment against their risks, being able to evaluate the individualized risk of future rupture in a patient with an UIA is a highly desired endpoint. Intense research efforts have, in turn, revealed factors associated with aneurysmal growth or rupture. Some depend on the patient characteristics (age, smoking, sex,...) and others on the aneurysm (size, regularity, location) [10, 11]. These variables have been synthesized in predictive scores for the risk of growth or rupture to help clinicians manage patients with UIA. They include most notably the PHASES (population, hypertension, age, size, earlier subarachnoid hemorrhage site), UCAS (Japanese Unruptured Cerebral Aneurysm Study), and ELAPSS scores (earlier subarachnoid hemorrhage, location of the aneurysm, age > 60 years, population, size of the aneurysm, shape of the aneurysm) [5, 12]. While these scores may be of interest to justify treatment in some cases, in our experience they are scantily used in daily practice due to both limited practicality, methodological issues in their elaboration, and perceived limited translation to real word complexity of care in patients with UIAs. Indeed, while they can be used to justify treatment in patients with a calculated “high risk” for future rupture, they can appear to be inadequately tuned to justify follow-up in patients with deemed lower risk for future rupture [13–16]. The proportion of patients with ruptured aneurysms but scoring low using the above-mentioned scales remains poorly explored.

In a large prospective multicentric European cohort, we aimed to evaluate whether the PHASES, UCAS, and ELAPSS scores in patients with aneurysmal subarachnoid hemorrhage would have predicted a high risk of aneurysmal rupture or growth.

Methods

Study sample

Academic centers treating patients with intracranial aneurysms were invited in the last semester of 2020 to

prospectively collect de-identified data through the JENI-SFNR network (Jeunes en Neuroradiologie Interventionnelle — Société Française de Neuroradiologie) [17], a trainee-led research collaborative. Each responding center was provided with an electronic case record form (CRF), and asked to prospectively include all patients admitted at their respective institution for a subarachnoid hemorrhage related to intracranial aneurysmal rupture between January 1 and March 31, 2021. Patients with post-traumatic or peri-mesencephalic subarachnoid hemorrhage with no causal intracranial aneurysm identified or patients not able to benefit from intracranial vessels' imaging were not included.

Data collection

The CRF collected all the elements of the PHASES, ELAPSS, and UCAS scores. Aneurysm characteristics (largest size, regularity, location) were assessed by each site investigator on cerebral angiograms, whether diagnostic or therapeutic. For patients with multiple aneurysms, the investigator pragmatically identified the acutely ruptured lesion, based on its appearance, the preferential location of bleed, and prior experience.

Data analysis, prediction scores, and threshold definitions

The PHASES, UCAS, and ELAPSS scores were calculated for all the included patients and descriptive analyses conducted using appropriate reports per variable type.

The PHASES score estimates a 5-year risk of aneurysmal rupture. In line with previous literature, we defined a very low risk of rupture as a 5-year risk < 0.4%, a low risk between 0.4 and 1.3%, an intermediate risk between 1.3 and 4.3%, and a high risk when > 5.3% [12, 14].

The UCAS score estimates a 3-year probability of aneurysmal rupture. Increments in the UCAS score have been associated with a risk of rupture < 1% for grade I, between 1 and 3% for grade II, between 3 and 9% for grade III, and above 9% for grade IV [5].

The ELAPSS score estimates predicted 3-year and 5-year risks of aneurysm growth. Low risk of growth has been defined at 5-year risk as < 13%, intermediate risk when between 13% and 19.3%, and high risk when > 19.3% [14, 18]. All these 3 scores and correlated risks are summarized in Supplemental data (Tables 2–7).

Ethics

A commitment to compliance was filed to the French National Information Science and Liberties Commission National for Healthcare Data before data centralization, in respect to the General Data Protection Regulation (CMR004, No. 2220301). Written informed consent was waived, but the patients or their proxies were informed that they could oppose the use of their de-identified health-related data for the purpose of research.

Results

Study sample and patient's characteristics

A total of 319 patients with aneurysmal subarachnoid hemorrhage were included at 17 centers (16 in France, and 1 in Switzerland).

Age

The patients were 56.5 ± 13.4 years old. Two-hundred and eleven patients (66%) were 60 years old or younger, and 262 patients (82%) were under 70 years. The age distribution is shown in Fig. 1.

Gender

In our cohort, 196 patients were women (61%).

History of SAH

Only 11 patients (3%) had a personal history of aneurysmal subarachnoid hemorrhage (SAH).

Blood pressure

One hundred and twenty patients (38%) were followed for, or had a history of, high blood pressure.

Nationality

No patients were of Finnish or Japanese ascent.

Aneurysm characteristics

Largest aneurysm size

The mean aneurysm largest diameter was 7 ± 3.7 mm, and the median size was 6 mm. Eighty-seven aneurysms (27%) were less than 5 mm; 183 aneurysms (57%) were less than 7 mm; and 5 aneurysms (2%) were measured more than

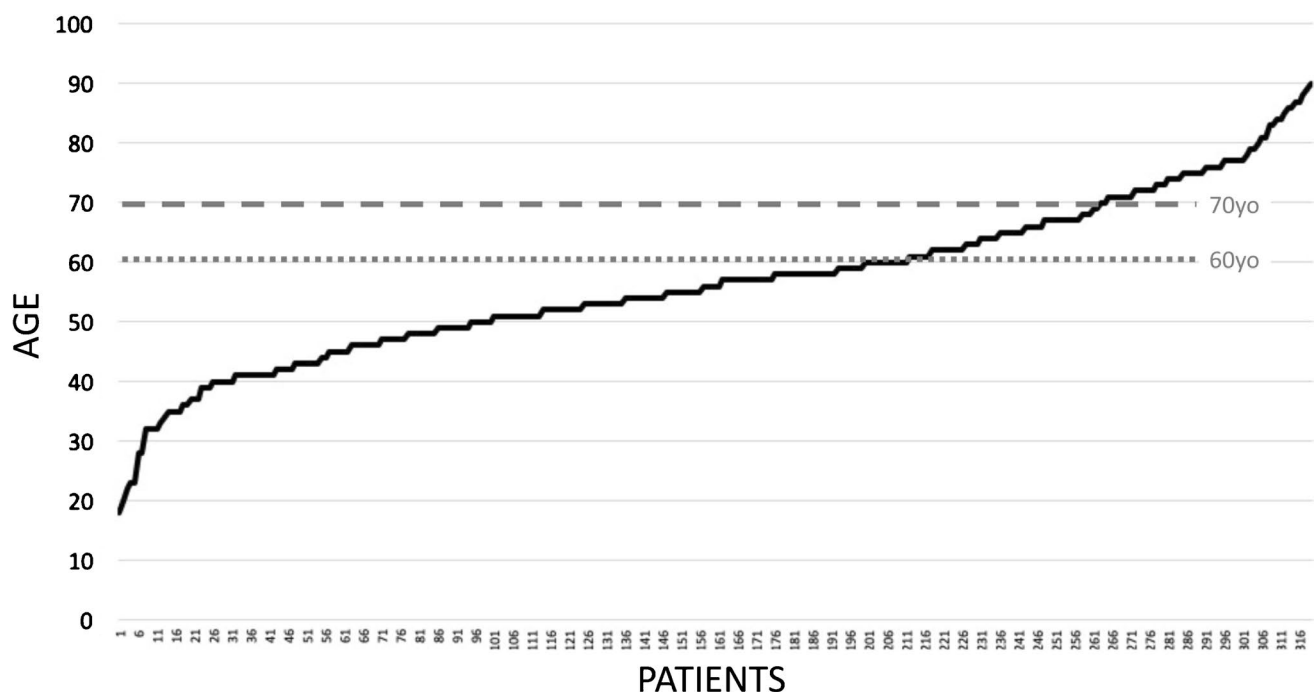


Fig. 1 Distribution of patients' age in the cohort. The lines represent the thresholds for scoring points in the ELAPSS (60 yo, dotted) and PHASES and UCAS (70 yo, dashed) scores. This graph is a visual

illustration of the important proportion of patients not meeting these criteria (below the lines) in our prospective sample

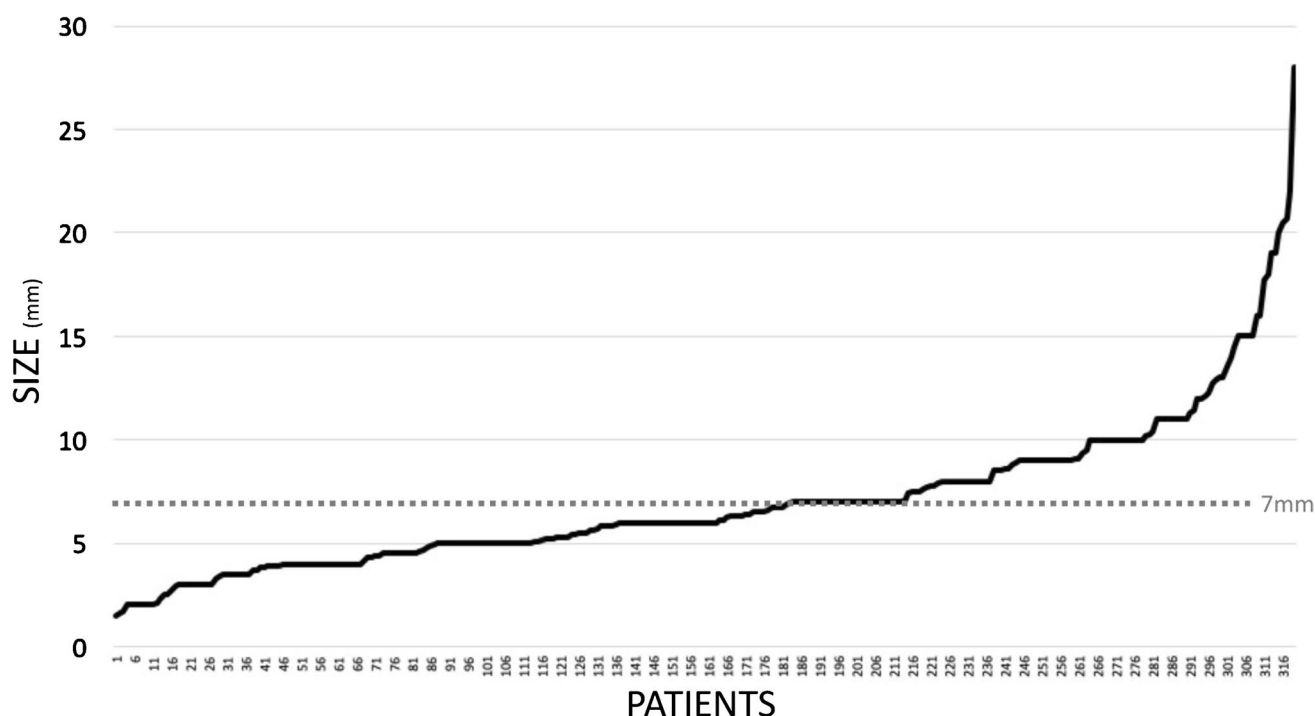


Fig. 2 Distribution of ruptured aneurysm sizes in the study sample. The dotted line (7 mm) is the minimal threshold for scoring points in the PHASES and UCAS scores

20 mm. The distribution of aneurysm size is represented in Fig. 2.

Locations

The majority of aneurysms were located on the anterior communicating artery (Acom, $n = 131$, 41%); 23 aneurysms (7%) were located at the anterior cerebral artery (ACA), 14 (4%) at the basilar artery (BA), 20 (6%) at the internal carotid artery (ICA), 62 (19%) at the middle cerebral artery (MCA), 47 (15%) at the posterior communicating artery (PCom), 15 (5%) at the vertebral artery (VA), and 7 (2%) at the posterior cerebral artery (PCA). See details of aneurysm locations in Table 1.

Irregularity

Two hundred and forty aneurysms (75%) had irregularities such as blebs or daughter cysts.

Multiple aneurysms Eighty-three patients (26%) had multiple aneurysms.

Patient and aneurysms' characteristics are summarized in Table 1.

PHASES, UCAS, and ELAPSS score distributions

The calculated average scores were 5.7 ± 2.6 , 6.1 ± 2.7 , and 17.3 ± 7.4 , respectively, for PHASES, UCAS, and ELAPSS scores.

One hundred eighty-four patients (57%) were classified as having a very low or low risk of rupture, according to the PHASES score; and 103 patients (32%) were classified in the first risk class of rupture risk using UCAS. Fifty-eight patients (18%) were classified as having a low risk of aneurysmal growth using ELAPSS. Only forty patients (13%) were classified as having a high risk of rupture with PHASES, and 114 patients (36%) were classified in the 3rd and 4th categories of rupture risk using UCAS. One hundred ninety-two patients (60%) were classified as high or very high risk of aneurysmal growth with ELAPSS. Figure 3 details the histogram distribution of each score and risk category.

Discussion

In a large prospective cohort of patients with ruptured intracranial aneurysms, we showed that an important proportion of the sample would have been classified as having a low to

Table 1 Baseline characteristics of included patients and aneurysms

Patients (<i>n</i> = 319)	<i>n</i> (%)
Female	196 (61%)
Age (years)	
≤ 60	211 (66%)
> 60	108 (34%)
< 70	262 (82%)
≥ 70	57 (18%)
Hypertension	120 (38%)
History of ASH	11 (3%)
Aneurysm largest size (mm)	7 ± 3.7
< 5	87 (27%)
< 7	183 (57%)
< 10	263 (82%)
< 20	314 (98%)
Aneurysm locations	
Anterior circulation	236 (74%)
ACA	23 (7%)
ACom	131 (41%)
ICA	20 (6%)
MCA	62 (19%)
Posterior circulation	83 (26%)
PCom	47 (15%)
PCA	7 (2%)
VA	15 (5%)
BA	14 (4%)
Aneurysm morphology	
Regular	79 (25%)
Irregular	240 (75%)

Variables are displayed as absolute number (percentage of column's total)

ACA anterior cerebral artery, ACom anterior communicating artery, BA basilar artery, ICA internal carotid artery, MCA middle cerebral artery, PCom posterior communicating artery, PCA posterior cerebral artery, VA vertebral artery

very low risk of rupture before the event (from 32 to 57%), if assessed using the PHASES and UCAS scores and as having a low to intermediate risk of growth using the ELAPSS score.

These prospectively acquired data question the use of risk assessment tools for intervention-related decision-making in patients with UIAs. Indeed, with 57% of ruptured aneurysms classified as very low or low risk of rupture, we can imagine that, if calculated before the rupture, this may have induced clinicians to an inappropriate strategy and prevent some patients from being considered for intervention. Furthermore, up to 32% of patients were classified with the UCAS scale as having a calculated risk lower than 1% rupture risk at 3 years. Building on previous literature, our data concur with the works by Pagiola et al. that showed that 79% of patients with ruptured intracranial aneurysms had a predicted risk of rupture at 5 years under 2%, using the PHASES score [13]. Similarly, Sturiale et al. retrospectively tested the 3 scores in patients admitted from 2015 to 2019 at two centers with ruptured aneurysms [14] and demonstrated that the majority of their sample also had risk scores in the lower ranges (from 28 to 46% of the population with PHASES and UCAS scores, respectively). Of note, both studies showed that the UCAS score classified proportionally less patients in the low-risk categories than PHASES or ELAPS. Still, in UCAS, the history of rupture from another aneurysm or the ethnic origin of the subject were not taken into account, whereas these 2 variables have been associated with varying over-risk in large samples. In our population, nonetheless, these 2 criteria also appeared to be of little relevance, since no patient were of Japanese or Finnish ascent, and since only 3% of patients had a history of SAH. At the individual level though, many have questioned the pertinence of this group-level associations, given the lack of true natural history data (that is,

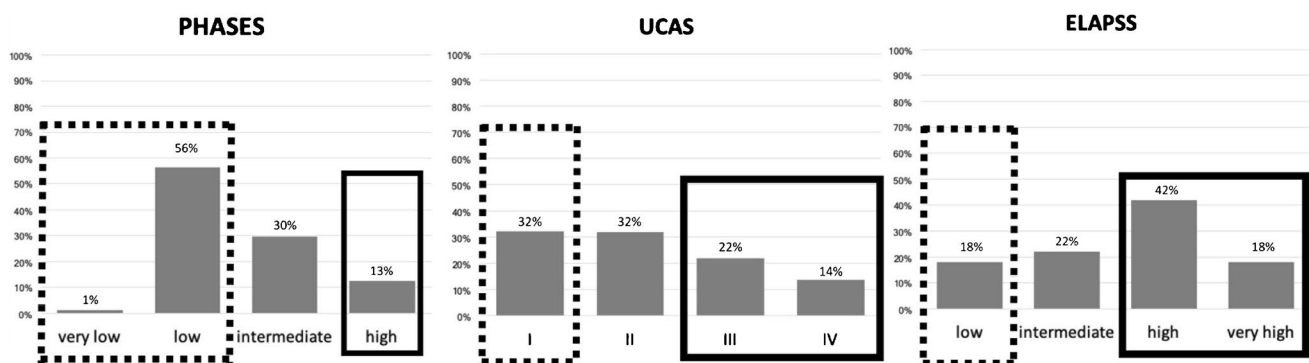


Fig. 3 PHASES, UCAS and ELAPSS score risk distribution in study sample. Dotted squares represent a low or very low risk and continued squares represent a high or very high risk

the aneurysm's course in the absence of treatment even in patients deemed at higher risk, given the obvious unethical assertions) [19]. The PHASES score was developed on the basis of a systematic review and patient level meta-analysis of 6 "natural history" cohorts totalizing 8382 patients with UIAs between 1956 and 2006 in Japan, the USA, Canada, and Europe, and including the cohort used to develop the UCAS score [12]. Fueling the debate of the absence of true natural history data in the literature, it is important to note that in these cohorts, 3346 patients were treated before inclusion; 3269 were treated during follow-up; and 1548 were lost to follow-up, truncating by half the true sample under longitudinal follow-up, and likely creating important biases in risk assessment. There have been, in turn, many criticisms of the PHASES score, including the failure to demonstrate an association between commonly accepted factors, such as smoking, female sex, the presence of wall irregularity or underlying collagenopathy, and a future rupture risk. The various external validation studies have, furthermore, often shown disappointing results, with an underestimation of the risk of rupture [13, 14, 20].

The UCAS score study-based population (5720 patients) is exclusively composed of patients of Japanese ascent, which explains why nationality was not taken into account in this score [5]. It consists of patients over 20 years of age with aneurysms of at least 3 mm. The authors cite as one of the main limitations of their study, the preventive treatment of more than 2000 small unruptured aneurysms, which were, therefore, subsequently excluded from the analysis. The deemed higher incidence of SAH in Japanese ascent for a similar incidence of UIAs [1] also questions the applicability of this score to other populations, and for instance mostly Caucasian European patients. As mentioned previously, the UCAS cohort is one of the 6 cohorts included in the pooled sample used to build the PHASES score, in which it represents 68% of the total number of patients, and explaining possible similarities between these 2 scores, as well as shared limitations. UCAS does not consider a previous history of SAH, as being a risk factor for rupture. In our study, this criterion has limited relevance since only 3% of patients had a previous history of SAH. Although, if the latter point is widely recognized as a risk factor for aneurysmal rupture, the monitoring and preventive treatment of aneurysms in patients with a history of aneurysmal rupture appears to reduce the incidence of bleeding from a distinct aneurysm in our study sample, likely biasing the results by indication. Indeed, several scholarly societies including the American Stroke Association encourage the systematic treatment of synchronous or metachronous aneurysms discovered in a patient with a history of SAH [4].

The application of measurements after aneurysmal rupture could constitute one limitation of our study. With an

average largest size of 6.97 mm, 57% of all included aneurysms were under the 7-mm cut-off identified by PHASES and UCAS, and only 6% were under the 3-mm cut-off of ELAPSS score. This data is in phase with a recent article published by Aubertin et al. which reported 3 subarachnoid hemorrhage in a followed cohort of 671 UIAs; the 3 ruptured aneurysms had a size inferior to 5 mm [21]. However, it is well known that upon rupture, aneurysms exhibit changes such as an increase in size and the appearance of wall irregularities or daughter cysts. These phenomena therefore probably artificially increase the scores within our ruptured aneurysm cohort compared to a prospective application of the pre-rupture scores and represent a systematic bias. Yet, this likely bias reinforces the results of the study, as if we had been able to measure the aneurysms before rupture, it would have decreased the calculated scores further.

Theoretically, a test aimed at predicting a rare but devastating event such as an aneurysm rupture needs to be very sensitive to prevent false negatives. Conversely, an important loss of specificity would mean that many patients would receive an unnecessary intervention, exposing them to potential treatment-related complications that may outweigh the benefits of treatment. While our work was not designed to answer the question of the overall adequacy of these tests, but only to provide a truncated estimation of their performances in a specific population, we must emphasize that a large proportion of patients with a ruptured aneurysm scored low on the 3 most used scores to predict rupture or growth. In turn, the proportion of false negatives is likely very high when considering the initial pool of patients, questioning the use of such scores for risk prediction in patients with UIAs. A pragmatic study, using long-term follow-up of patients after UIA diagnosis, is likely the only way forward to optimize treatment decision. Given the stakes, care trials may represent the only viable option to further our understanding of UIA management [15].

The absence of centralized core-lab reading is a limitation of our work. However, all centers that participated in the study are expert centers in the management of ruptured aneurysms, with extensive experience in this pathology, suggesting very good reliability of otherwise simple and reproducible measurements. Moreover, Takao et al. reported that manual aneurysm measurements are most accurate with DSA than computed tomography or magnetic resonance imaging; the reason why all patients admitted in neuroradiology units without cerebral angiogram permitting aneurysmal sizing were excluded of our analysis [22].

In our sample, we found that a quarter of our patients with subarachnoid hemorrhage had multiple intracranial aneurysms. It is coherent with Kaminogo et al. which reported that multiple intracranial aneurysms are discovered in 15 to 35% of patients with subarachnoid hemorrhage [23].

One of the strengths of our study is the prospective identification of consecutive cases. Indeed, all the adult patients with SAH presenting in the different centers, meeting the inclusion criteria, were included in the study. A second strength is represented by a massively multicentric design with 17 participating centers, likely increasing the external validity of our findings to similarly selected European, mostly Caucasian patients.

Conclusion

This prospective study showed that in a European sample of patients with aneurysmal subarachnoid hemorrhage, the rupture risk would have been estimated to be low to very low for most patients using 3 common risk-assessment tools, questioning their use for decision-making in patients with UIAs.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00234-022-02987-w>.

Acknowledgements

Study Group collaborators
Stéphane Velasco¹, Victor Dumas¹, Ayoub Guerrab¹, Samy Boucebc¹, Charbel Mounayer², Susanna Saleme², Jérôme Berge³, Xavier Barreau³, Patrice Ménégon³, Thomas Toudias³, Jean Papaxanthos³, Mourad Cheddad⁴, El Aouni⁴, Nader Sourour⁵, Kevin Premat⁵, Stéphanie Lenck⁵, Bertrand Mathon⁵, Anne-Laure Boch⁵, Vincent Degos⁵, Alice Jacquens⁵, Marios Psychogios⁶, Emmanuel Gerardin⁷, Chrysanthi Papagiannaki⁷, Adelya Curado⁷, Margaux Lefebvre⁷, Hubert Desal⁸, Lili Detraz⁸, Benjamin Daumas-Duport⁸, Pierre-Louis Alexandre⁸, Cédric Lenoble⁸, Elisabeth Auffray-Calvier⁸, Jesus Aguilar-Garcia⁸, Alina Linti-Gaultier⁸, Rémy Beaujeux⁹, Petru Coltea⁹, Anca Hasiu⁹, Monica Manisor⁹, Olivier Naggara¹⁰, Denis Trystram¹⁰, Christine Rodriguez-Régent¹⁰, Quentin Alias¹⁰, Hervé Brunel¹¹, Anthony Reyre¹¹, Xavier Carle¹¹, Pierre Lehmann¹¹, Philippe Dory-Lautrec¹¹, Christophe Cognard¹², Fabrice Bonneville¹², Anne Christine Januel¹², Philippe Tall¹², Anne Pasco Papon¹³, Marina Sachet¹⁴, François Vassal¹⁴, Marie-Charlotte Tetard¹⁴, René Anxionnat¹⁵, Benjamin Gory¹⁵, Liang Liao¹⁵, Anne Laure Derelle¹⁵, Patricio Muszynski¹⁵

¹University Hospital of Poitiers, Poitiers, France

²University Hospital of Limoges, Limoges, France

³University Hospital of Bordeaux, Bordeaux, France

⁴University Hospital of Brest, Brest, France

⁵Pitié Salpêtrière Hospital, Paris, France

⁶University Hospital of Basel, Basel, Switzerland

⁷University Hospital of Rouen, Rouen, France

⁸University Hospital of Nantes, Nantes, France

⁹University Hospital of Strasbourg, Strasbourg, France

¹⁰Saint-Anne Hospital, Paris, France

¹¹University Hospital of Marseille, Marseille, France

¹²University Hospital of Toulouse, Toulouse, France

¹³University Hospital of Angers, Angers, France

¹⁴University Hospital of Saint-Etienne, Saint-Etienne, France

¹⁵University Hospital of Nancy, Nancy, France

Author contribution P. Lognon: conceptualization, writing — original draft preparation.

F. Gariel: reviewing and editing.

G. Marnat: reviewing and editing.

J. Darcourt: reviewing and editing.

P. Constant dit Beaufils: reviewing and editing.

J. Burel: reviewing and editing.

E. Shotar: reviewing and editing.

J. F. Hak: reviewing and editing.

C. Fauché: reviewing and editing.

B. Kerleroux: reviewing and editing.

A. Guédon: reviewing and editing.

J. Ognard: reviewing and editing.

G. Forestier: reviewing and editing.

R. Pop: reviewing and editing.

C. Paya: reviewing and editing.

J. B. Veyrières: reviewing and editing.

P. Sporns: reviewing and editing.

J. B. Giro: reviewing and editing.

R. Zannoni: reviewing and editing.

F. Zhu: reviewing and editing.

A. Crespy: reviewing and editing.

V. L'Allinec: reviewing and editing.

D. Mihoc: reviewing and editing.

A. Rouchaud: reviewing and editing.

J. C. Gentric: reviewing and editing.

W. Ben Hassen: reviewing and editing.

N. Raynaud: reviewing and editing.

B. Testud: reviewing and editing.

F. Clarençon: reviewing and editing.

B. Kaczmarek: reviewing and editing.

R. Bourcier: reviewing and editing.

G. Bellanger: reviewing and editing.

G. Boulouis: writing — reviewing and editing.

K. Janot: conceptualization, writing — reviewing and editing.

Funding information This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Code availability NA.

Compliance with ethical standards

Conflict of interest We declare that we have no conflict of interest.

Ethical approval As for all non-interventional studies of de-identified data in France, a commitment to compliance was filed to the French National Information Science and Liberties Commission National for Healthcare Data before data centralization, in respect to the General Data Protection Regulation (Reference Methodology CMR004, No. 2220301).

Informed consent Written informed consent was waived, but patients or their proxies were informed that they could oppose the use of their de-identified health-related data for the purpose of research and its publication.

Data sharing Data are available upon a reasonable request.

References

1. Vlak MH, Algra A, Brandenburg R, Rinkel GJ (2011) Prevalence of unruptured intracranial aneurysms, with emphasis on sex, age, comorbidity, country, and time period: a systematic review and meta-analysis. *Lancet Neurol* 10:626–636. [https://doi.org/10.1016/S1474-4422\(11\)70109-0](https://doi.org/10.1016/S1474-4422(11)70109-0)

2. Rinkel GJE, Djibuti M, Algra A, van Gijn J (1998) Prevalence and risk of rupture of intracranial aneurysms. *Stroke* 29:251–256. <https://doi.org/10.1161/01.STR.29.1.251>
3. Komotar RJ, Mocco J, Solomon RA (2008) Guidelines for the surgical treatment of unruptured intracranial aneurysms - controversies in the management of cerebral aneurysms. *Neurosurgery* 62:183–194. <https://doi.org/10.1227/01.NEU.0000311076.64109.2E>
4. Gregory TB, Brown RD, Sepideh A-H et al (2015) Guidelines for the management of patients with unruptured intracranial aneurysms. *Stroke* 46:2368–2400. <https://doi.org/10.1161/STR.0000000000000070>
5. Tominari S, Morita A, Ishibashi T et al (2015) Prediction model for 3-year rupture risk of unruptured cerebral aneurysms in Japanese patients: cerebral aneurysm rupture risk. *Ann Neurol* 77:1050–1059. <https://doi.org/10.1002/ana.24400>
6. Wermer MJH, van der Schaaf IC, Algra A, Rinkel GJE (2007) Risk of rupture of unruptured intracranial aneurysms in relation to patient and aneurysm characteristics: an updated meta-analysis. *Stroke* 38:1404–1410. <https://doi.org/10.1161/01.STR.0000260955.51401.cd>
7. D'Souza S (2015) Aneurysmal subarachnoid hemorrhage. *J Neurosurg Anesthesiol* 27:222–240. <https://doi.org/10.1097/ANA.0000000000000130>
8. Naggara ON, White PM, Guilbert F et al (2010) Endovascular treatment of intracranial unruptured aneurysms: systematic review and meta-analysis of the literature on safety and efficacy. *Radiology* 256:887–897. <https://doi.org/10.1148/radiol.10091982>
9. Kotowski M, Naggara O, Darsaut TE et al (2013) Safety and occlusion rates of surgical treatment of unruptured intracranial aneurysms: a systematic review and meta-analysis of the literature from 1990 to 2011. *J Neurol Neurosurg Psychiatry* 84:42–48. <https://doi.org/10.1136/jnnp-2011-302068>
10. Etminan N, Buchholz BA, Dreier R et al (2014) Cerebral aneurysms: formation, progression, and developmental chronology. *Transl Stroke Res* 5:167–173. <https://doi.org/10.1007/s12975-013-0294-x>
11. Nohra C, Hoh BL, David H (2013) Review of cerebral aneurysm formation, growth, and rupture. *Stroke* 44:3613–3622. <https://doi.org/10.1161/STROKEAHA.113.002390>
12. Greving JP, Wermer MJH, Brown RD et al (2014) Development of the PHASES score for prediction of risk of rupture of intracranial aneurysms: a pooled analysis of six prospective cohort studies. *Lancet Neurol* 13:59–66. [https://doi.org/10.1016/S1474-4422\(13\)70263-1](https://doi.org/10.1016/S1474-4422(13)70263-1)
13. Pagiola I, Mihalea C, Caroff J et al (2020) The PHASES score: to treat or not to treat? Retrospective evaluation of the risk of rupture of intracranial aneurysms in patients with aneurysmal subarachnoid hemorrhage. *J Neuroradiol* 47:349–352. <https://doi.org/10.1016/j.neurad.2019.06.003>
14. Sturiale CL, Stumpo V, Ricciardi L et al (2020) Retrospective application of risk scores to ruptured intracranial aneurysms: would they have predicted the risk of bleeding? *Neurosurg Rev*. <https://doi.org/10.1007/s10143-020-01352-w>
15. Darsaut TE, Desal H, Cognard C et al (2020) Comprehensive aneurysm management (CAM): an all-inclusive care trial for unruptured intracranial aneurysms. *World Neurosurg* 141:e770–e777. <https://doi.org/10.1016/j.wneu.2020.06.018>
16. Foreman PM, Hendrix P, Harrigan MR et al (2018) PHASES score applied to a prospective cohort of aneurysmal subarachnoid hemorrhage patients. *J Clin Neurosci* 53:69–73. <https://doi.org/10.1016/j.jocn.2018.04.014>
17. Kerleroux B, Bourcier R, Rouchaud A et al (2021) Interventional Neuroradiology Trainee-led Research Collaborative JENI, moving forward. *J Neuroradiol* 48:137–138. <https://doi.org/10.1016/j.neurad.2021.01.007>
18. Backes D, Rinkel GJE, Greving JP et al (2017) ELAPSS score for prediction of risk of growth of unruptured intracranial aneurysms. *Neurology* 88:1600–1606. <https://doi.org/10.1212/WNL.00000000000003865>
19. Naggara O, Darsaut T, Trystram D et al (2014) Unruptured intracranial aneurysms: why we must not perpetuate the impasse for another 25 years. *Lancet Neurol* 13:537–538. [https://doi.org/10.1016/S1474-4422\(14\)70091-2](https://doi.org/10.1016/S1474-4422(14)70091-2)
20. Hilditch CA, Brinjikji W, Tsang A et al (2021) Application of PHASES and ELAPSS scores to ruptured cerebral aneurysms: how many would have been conservatively managed. *J Neurosurg Sci* 65:33–37. <https://doi.org/10.23736/s0390-5616.18.04498-3>
21. Aubertin M, Jourdain C, Thépenier C et al (2021) Results of watchful waiting of unruptured intracranial aneurysms in a Western patient population: a single-center cohort. *J Neurointerv Surg:neurintsurg-2021–018151*. <https://doi.org/10.1136/neurintsurg-2021-018151>
22. Takao H, Murayama Y, Ishibashi T et al (2010) Comparing accuracy of cerebral aneurysm size measurements from three routine investigations: computed tomography, magnetic resonance imaging, and digital subtraction angiography. *Neurol Med Chir (Tokyo)* 50:893–899. <https://doi.org/10.2176/nmc.50.893>
23. Kaminogo M, Yonekura M, Shibata S (2003) Incidence and outcome of multiple intracranial aneurysms in a defined population. *Stroke* 34:16–21. <https://doi.org/10.1161/01.STR.0000046763.48330.AD>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

P. Lognon¹ · F. Gariel² · G. Marnat² · J. Darcourt³ · P. Constant dit Beaufils^{4,5} · J. Burel⁶ · E. Shotar⁷ · J. F. Hak⁸ · C. Fauché⁹ · B. Kerleroux¹⁰ · A. Guédon¹¹ · J. Ognard¹² · G. Forestier¹³ · R. Pop¹⁴ · C. Paya¹⁵ · J. B. Veyrières¹⁵ · P. Sporns^{16,17} · J. B. Girot¹⁸ · R. Zannoni¹⁹ · F. Zhu²⁰ · A. Crespy¹ · V. L'Allinec¹⁸ · D. Mihoc¹⁴ · A. Rouchaud¹³ · J. C. Gentric¹² · W. Ben Hassen¹⁰ · N. Raynaud⁹ · B. Testud⁸ · F. Clarençon⁷ · B. Kaczmarek⁶ · R. Bourcier^{4,5} · G. Bellanger³ · G. Boulouis¹ · Kevin Janot¹  · on behalf of the JENI Research Collaborative

¹ University Hospital of Tours, Tours, France

² University Hospital of Bordeaux, Bordeaux, France

³ University Hospital of Toulouse, Toulouse, France

⁴ L'institut du Thorax, University of Nantes, INSERM, CNRS, Nantes, France

⁵ University Hospital of Nantes, Nantes, France

⁶ University Hospital of Rouen, Rouen, France

⁷ Pitié Salpêtrière Hospital, Paris, France

⁸ University Hospital of Marseille, Marseille, France

⁹ University Hospital of Poitiers, Poitiers, France

¹⁰ Sainte-Anne Hospital, Paris, France

¹¹ Lariboisière Hospital, Paris, France

¹² University Hospital of Brest, Brest, France

¹³ University Hospital of Limoges, Limoges, France

¹⁴ University Hospital of Strasbourg, Strasbourg, France

¹⁵ University Hospital of Saint-Pierre, Saint-Pierre, La Réunion, France

¹⁶ University Hospital of Basel, Basel, Switzerland

¹⁷ University Medical Center of Hamburg-Eppendorf, Hamburg, Germany

¹⁸ University Hospital of Angers, Angers, France

¹⁹ University Hospital of Saint-Etienne, Saint-Etienne, France

²⁰ University Hospital of Nancy, Nancy, France